

The first sphere and the second sphere no longer collide head-on.

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(a)(i) Faulty Lamp: Lamp C

Nature of fault: Lamp is shorted(or short circuit).

(ii) Using a power supply might short Lamp A, high current pass through the circuit and would damage the power supply/lamps/blow fuse in supply.

(iii) $15\ \Omega$

(b)(i) Potential difference across wire is $V = IR$ where R is $\rho \frac{L}{A}$ where I is current in wire, R is resistance of wire, ρ is resistivity of wire, L is length of wire and A is cross-sectional area of wire. Since I , ρ and A are constant, therefore V is directly proportional to length of wire.

(ii) Potential across thermistor = pd across QM
 $= 1800/3000 (4.5) = 2.7\text{ V}$

$$\frac{L}{100} = \frac{2.7}{4.5}$$

$$L = 60.0 \text{ cm}$$

3(b)(ii) As temperature rises, thermistor resistance decreases. Pd across QM will be smaller, thus QM is shorter.

4. (a) (i)